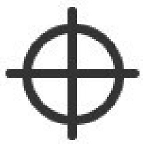
UNIT – III

Location, Runout and Profile Tolerances

Tolerances of Location :

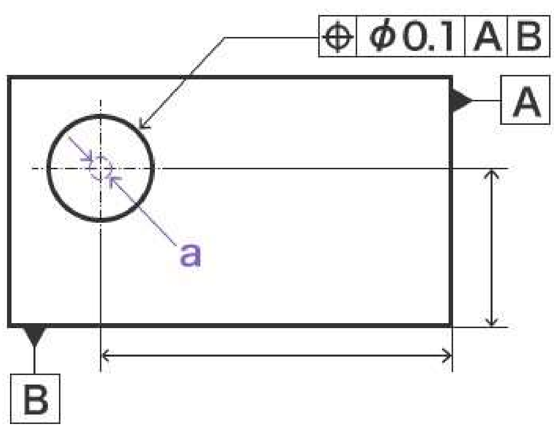
- Location tolerance determines the location (true position) of the feature in relation to a reference. A datum is always necessary to indicate location tolerance; as such, it is a geometric tolerance for features related to datums. Types of Tolerances of Loca on:

True position:



- The true position requirement specifies the accuracy of the position in relation to the datum (reference plane or line).

Sample indication:

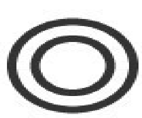


a) Within 0.1 mm dia.

Explanation of drawings:

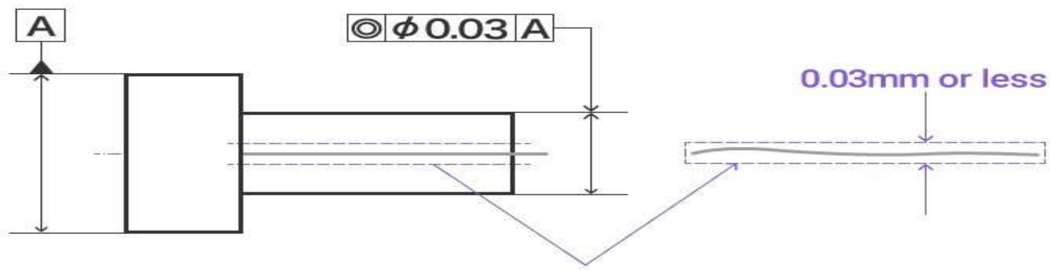
- The center of the circle indicated by the indication arrow must be within a circle with a diameter of 0.1 mm.

Coaxiality:



The coaxiality requirement specifies the coaxiality of the axes of two cylinders (no deviation of the central axis).

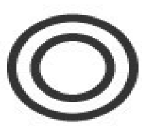
Sample indication:



Explanation of drawings:

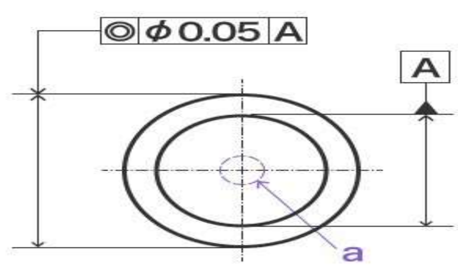
- The axis of the cylinder indicated by the indication arrow must be within a cylinder with datum axis line A as the axis and a diameter of 0.03 mm.

Concentricity:



- The concentricity requirement specifies the accuracy of concentricity of the axes of two cylinders (no deviation of the center). Unlike coaxiality, the datum is the center point (plane).

Sample indication:



a) Within a circle with 0.05 mm dia.

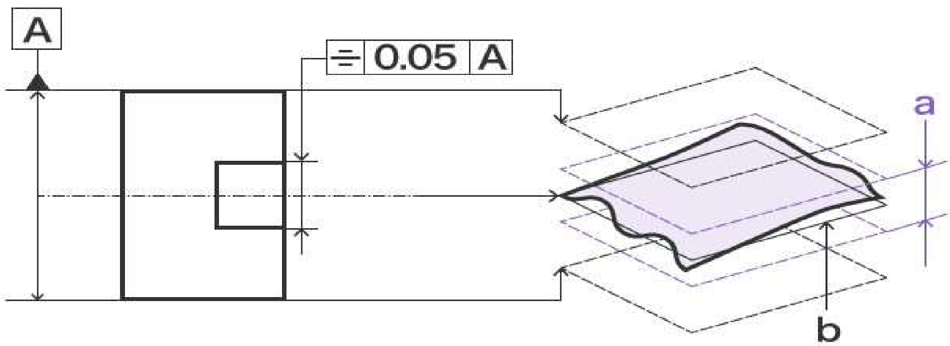
Explanation of drawings:

- The axis of the cylinder indicated by the indication arrow must be within a cylinder with datum axis line A as the axis and a diameter of 0.05 mm.

Symmetry:

The symmetry requirement specifies the accuracy of how symmetrical a target is to the datum (reference plane).

Sample indication:



a) Within 0.05 mm from the theoretical center plane
b) Theoretical center plane

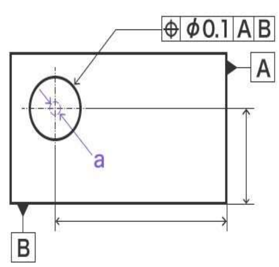
Explanation of drawings:

 The center plane indicated by the indication arrow must be between two parallel planes symmetrical to the datum center plane A and separated from each other by 0.05 mm.

Verifica on techniques for Loca on tolerance:

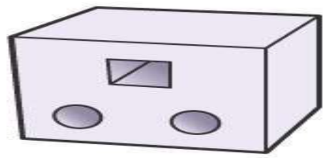
Measuring True posi on:

Sample Drawings:



a) Within 0.1 mm dia.

Using a Gauge:

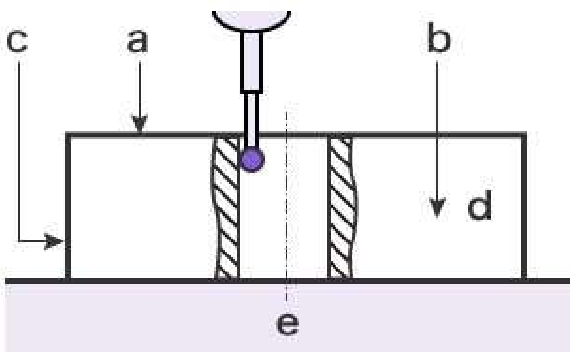


A pass/fail judgment is performed using a measuring gauge or inspection gauge. It has the benefit of having no variation in operation speed and inspection quality deriving from the skill level of the operator as well as supporting automation thanks to its simple operation.

DISADVANTAGES:

- Gauges need to be prepared for each target—custom-made gauges have a substantial up-front cost, which makes it difficult to introduce during the prototyping phase.

Using a Coordinate Measuring Machine (CMM):



a) X-Y plane, datum plane A
b) Z-X plane, datum plane B
c) Y-Z plane, datum plane C
d) Target
e) Surface plate

- Set the reference plane and put the stylus on the measurement point on the target. The measurement result is instantly displayed on the screen.

- Cartesian coordinates can also be measured, and composite true position can be output with a single measurement.
  To measure bores, take several measurements by changing the depth to output verification results of cylindricity, perpendicularity, and straightness.

Measuring coaxiality:

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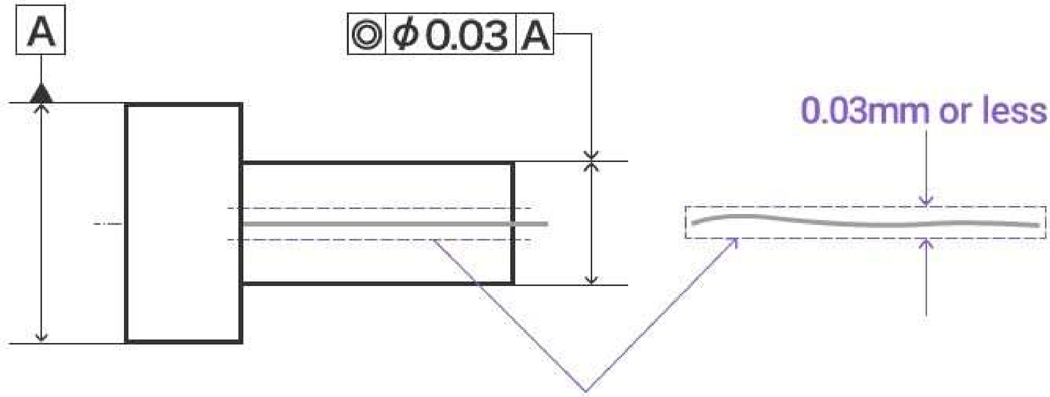
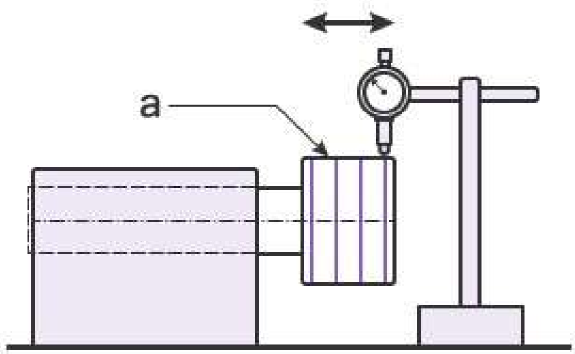
- When measuring coaxiality, you are checking for any deviation in the central axes (i.e., that the axes of two cylinders are coaxial).

Sample Drawings



Using a Dial Gauge



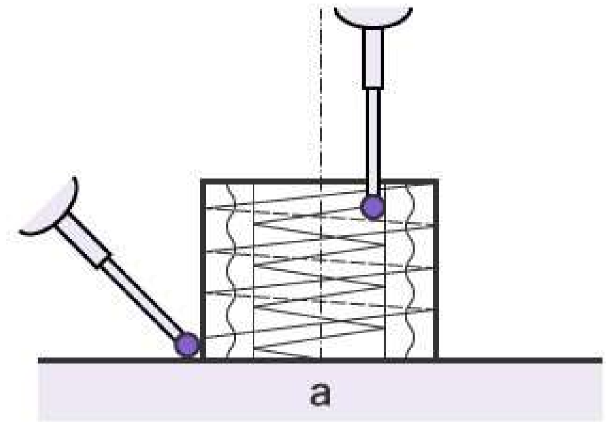
a) Any measurement loca on

- Hold the target in place and put the dial gauge on the vertex of the circumference for which tolerance is indicated.
  Rotate the target and measure the maximum and minimum run-out values using the dial gauge.
  Repeat measurements on the specified axis. The greatest maximum-minimum difference is used as the coaxiality.

DISADVANTAGES:

- Factors such as the angle and strength used to put the dial gauge on the target affect the measured value, which implies that measurements may differ depending on the operator.
  Because measurements are repeated on the specified axis, the measurements and their results must be checked at all times.

Using a Coordinate Measuring Machine (CMM)



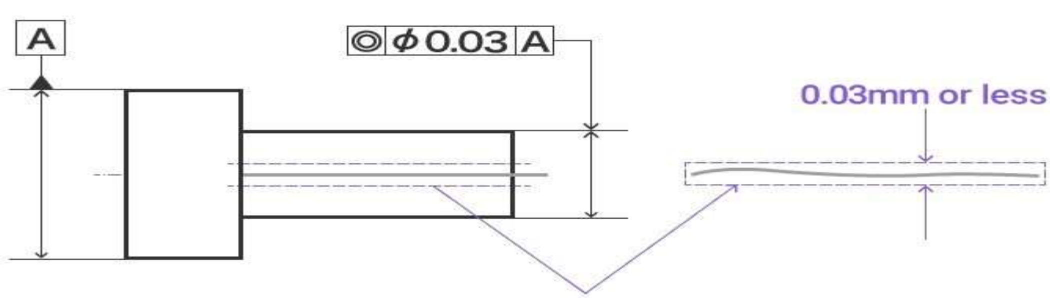
a) Surface plate

- Put the stylus on the measurement point on the datum element (cylinder), and then put the stylus on the measurement point on the target element (cylinder) to measure the coaxiality. The measurement result is recorded in the measuring machine. There are two modes of putting the stylus on a surface: point measurement, which measures by putting the stylus on the surface for each measurement; and auto trigger (scanning) measurement, which measures continuous points by moving the stylus while maintaining contact with the surface.
  This enables the spiral movement of the stylus to measure cylinders, which are difficult for the stylus to come into contact with.

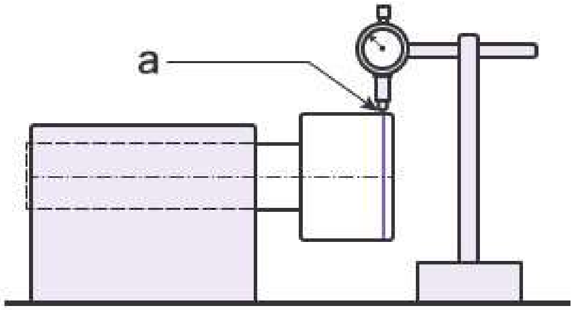
Measuring concentricity:

- When measuring [concentricity](), you are checking the accuracy of the coaxiality of the axes of two cylinders, that the center points match. Unlike coaxiality, the datum is the center point (plane).

Sample Drawings



Using a Dial Gauge



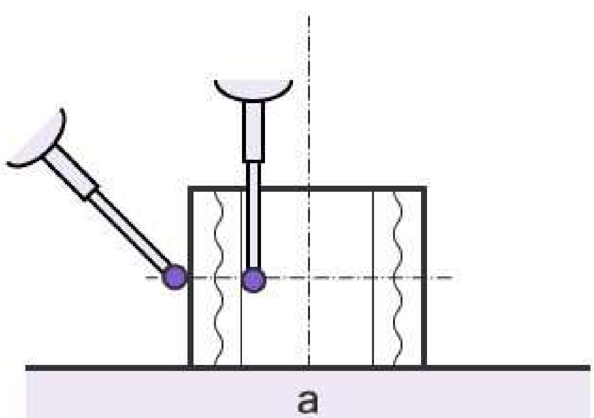
a) Specified measurement posi on

Hold the target in place and put the dial gauge on the vertex of the circumference for the axis for which tolerance is indicated. Rotate the target and measure the maximum and minimum run-out values using the dial gauge. Measure around the specified circumference. The greatest maximum-minimum difference is used as the concentricity.

DISADVANTAGES:

- Factors such as the angle and strength used to put the dial gauge on the target affect the measured value, which implies that measurements may differ depending on the operator.
  The friction between the tip of the dial gauge and the surface of the target may also leave scratches on the surface of the target.

Using a Coordinate Measuring Machine (CMM)



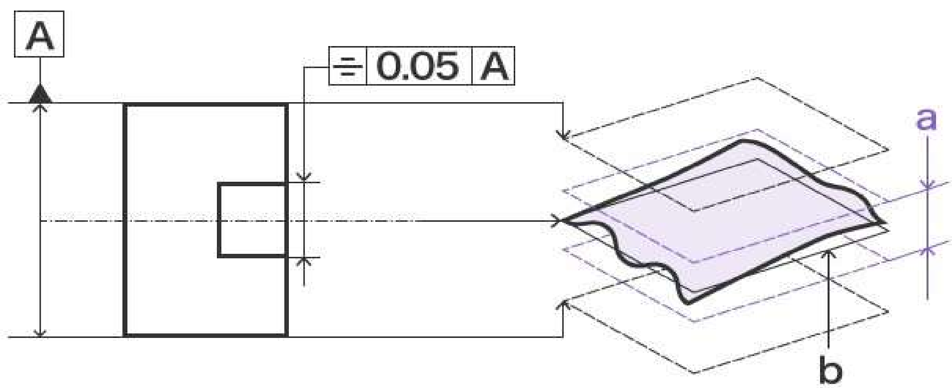
a) Surface plate

Unlike with coaxiality, you measure the circle of the plane.
Put the stylus on the measurement point on the datum circle, and then put the stylus on the measurement point on the target circle to measure the concentricity. The stylus only comes into light contact with the surface and does not scratch the target.

Measuring symmetry:

- When measuring symmetry, you are checking how accurately symmetrical the target is against the datum (reference plane).

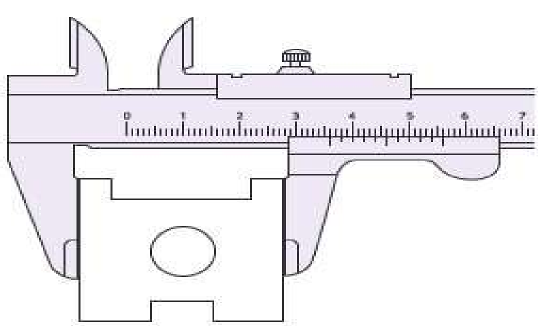
Sample Drawings



a) Within 0.05 mm from the theore cal center plane

b) Theore cal center plane

Using a Caliper or Micrometer



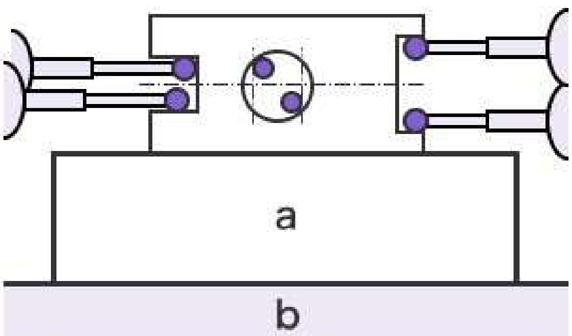
- Measure parts of the target using an analog caliper or micrometer to check the symmetry.

It is useful for repeated measurements of single items thanks to its simple, quick usability.
Both calipers and micrometers come in various types, which are selectively used depending on the location and form to be measured.

DISADVANTAGES:

- The accuracy of measured values and speed of measurement rely on the skill level of the operator in addition to the measurement error of the individual instrument. Furthermore, while size can be measured as these instruments measure the length between two points, geometric tolerance (form) is difficult to measure.
Additionally, the measurement data needs to be handwritten for recordkeeping.

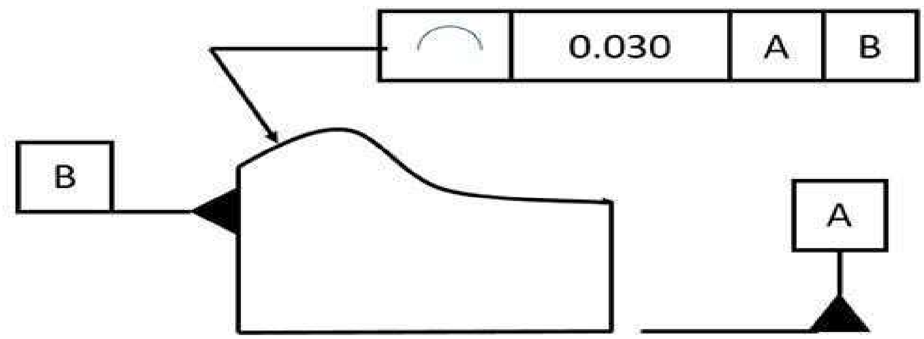
Using a Coordinate Measuring Machine (CMM)



a) Stage
c) Surface plate

- The reference element (plane) set-up and the deviation from the target element (plane) can be quickly and accurately measured by anyone just by putting the stylus on each measurement point.
The measurement result is recorded in the measuring machine.

Tolerances of profiles of lines and surfaces with or without datums:

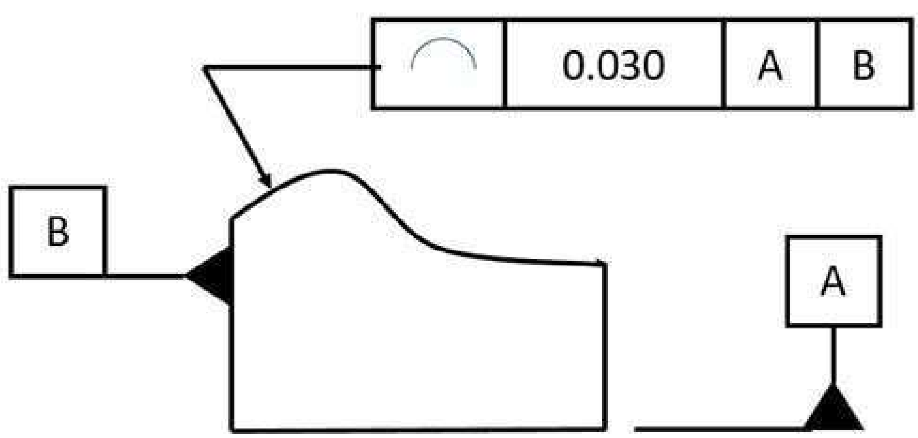


Any part is composed of planes and surfaces. The measurement and evaluation of surface shape error is also a very important item in product inspection. In the mechanical manufacturing industry, the profile index is used to evaluate the error.

- In this article, we will look at the profile of line tolerance and the profile of surface belonging to the profile group. Profile of a Line is very common, it helps us to manufacture parts with complex shapes. You will know the symbols, definitions, tolerance zones and precautions of Line profile GD&T Symbol:]

Profile of a Line in GD&T

- The profile of a line tolerance is a 2D GD&T dimension that defines and controls linear/curve features or surface cross-sections within specified limits. We can use it to control the form, position, direction and size of features. This dimension can only be used for surfaces, not for locating central axes or planes.



- The line profile callout is usually used in the case of complex curves. These can be simple or advanced algebraic curves. If it is called on the surface, it is just like the radius on the part – the contour of the line will specify how much the cross section changes from the true bending radius. The profile of a line intercepts a cross section at any point along the surface and sets a tolerance zone on either side of the profile.

- Line profiles are typically applied to parts with different cross sections, or to specific cross sections that are critical to part functionality.

Marking of Line Profile Tolerance and Meaning of Tolerance Zone:

- Line profile tolerance zone Line profile tolerance creates a true profile where the curve should ideally be located. The actual curve is evaluated based on this true profile.

- The tolerance zone consists of two parallel lines that follow the true contour in either direction.

- The distance between two parallel curves is the tolerance limit. Each point on the actual curve must lie between these two lines of the tolerance zone for approval. The application to which the tolerance zone applies can be specified on the drawing. The tolerance zone may or may not be referenced by the datum.

No benchmark requirement

- The tolerance zone is the area between two enveloping lines of a series of circles with diameter of tolerance value t, and the center of each circle is located on the ideal contour line.

Having benchmark requirements

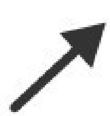
- The tolerance zone is the area bounded by two envelopes of a series of circles whose diameter is equal to the tolerance value t and whose center is located on the theoretically correct geometric shape of the measured feature determined by datum plane A and datum plane B.

- Remarks: When there is no datum requirement for line profile tolerance, it is a form tolerance, and when there is datum requirement, it is a position tolerance.

Tolerances of Runout:

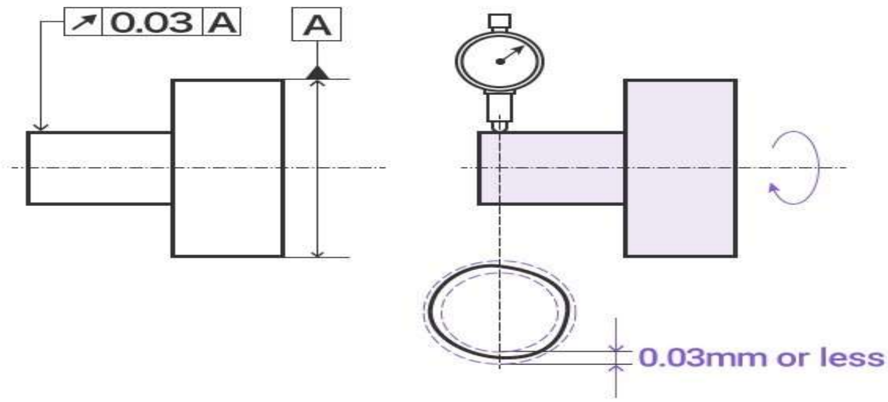
- Run-out tolerance is a geometric tolerance that specifies the run-out fluctuation of a target's feature when the target (part) is rotated on an axis (specified straight line). A datum is always necessary to indicate run-out tolerance; as such, it is a geometric tolerance for features related to datums.

Circular Run-out:



Circular run-out specifies the run-out of any part of a circumference when a part is rotated.
To meet the circular run-out requirement, the run-out of the measured value when the part is rotated must be within the specified range.

Sample indication:



Explanation of drawings:

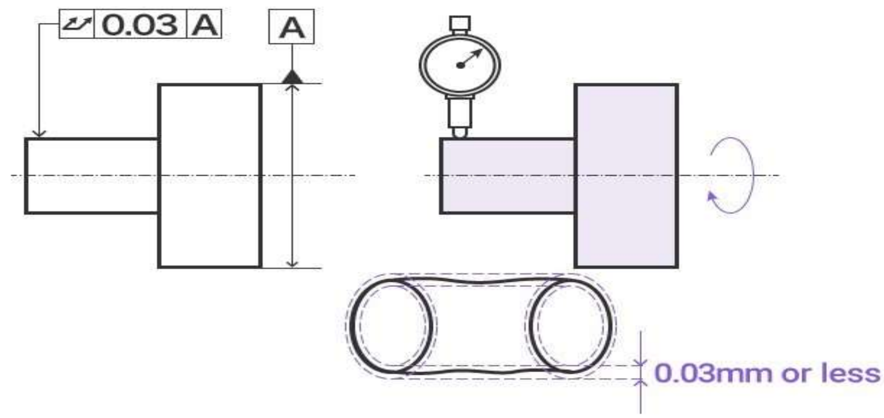
- The run-out of the cylindrical surface in the radial direction, indicated by the indication arrow, must not surpass 0.03 mm on any measurement plane perpendicular to the datum axis line when the target is rotated once on the datum axis line.

Total Run-out:

- Total run-out specifies the run-out of the entire surface of a part when it is rotated. To meet the total run-out requirement, the run-out of the measured value of the entire cylinder surface must be within the specified range.

Sample indication:

□



Explanation of drawings

- The total run-out of the cylindrical surface in the radial direction, indicated by the indication arrow, must not surpass 0.03 mm at any point on the cylindrical surface when the cylinder part is rotated on the datum axis line.

Tolerancing of Angles and cones:

Tolerancing of Angles:

Angularity:

Description:

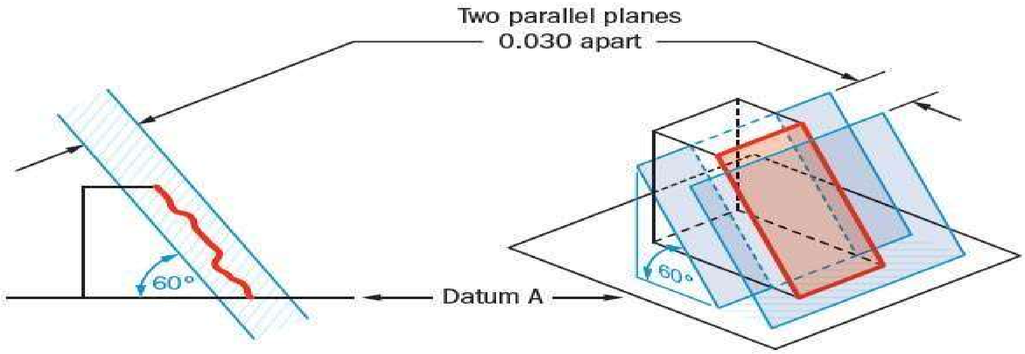
- Angularity is the symbol that describes the specific orientation of one feature to another at a referenced angle. It can reference a 2D line referenced to another 2D element, but more commonly it relates the orientation of one surface plane relative to another datum plane in a 3Dimensional tolerance zone. The tolerance does not directly control the angle variation and should not be confused with an angular dimension tolerance such as ± 5°. In fact, the angle, for now, becomes a Basic Dimension, since it is controlled by your geometric tolerance. The tolerance indirectly controls the angle by controlling where the surface can lie based on the datum. See the tolerance zone below for more details.

- Maximum material condition or axis control can also be called out for angularity although the use in design and fabrication is very uncommon since gauging a hole or pin at an angle is difficult. When angularity is called out on an axis, the tolerance zone now becomes a cylinder around the referenced axis at an angle to the datum. The page on Perpendicularity goes into this type of reference in further detail since it is more common with perpendicularity.

GD&T Tolerance Zone:

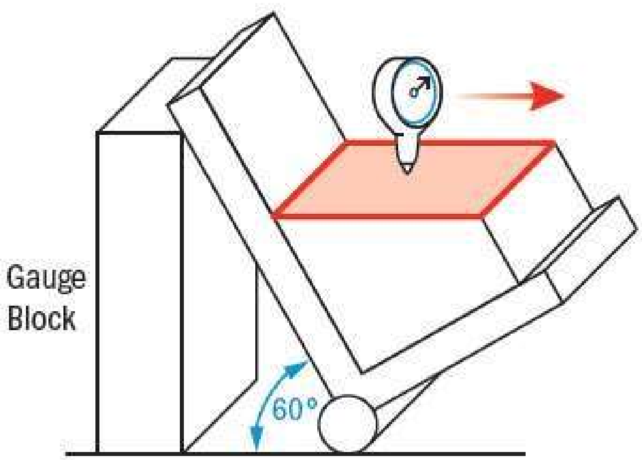
- Two parallel planes or lines which are oriented at the specified angle in relation to a datum. All points on the referenced surface must fall into this tolerance zone.

- Angularity does not directly control the angle of the referenced surface; it controls the envelope (like flatness) that the entire surface can lie.



Gauging / Measurement:

- Angularity is measured by constraining a part, usually with a sine bar, tilted to the reference angle, so that the reference surface is now parallel to the granite slab. By setting the part at an angle the flatness can now be measured across the now horizontal reference surface. The entire variation must not fall outside the tolerance zone.

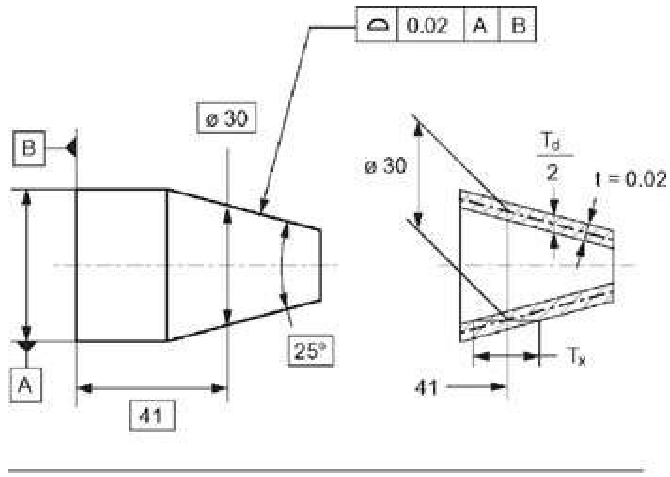


Tolerancing of Cones:

With cones, the following characteristics may be toleranced:

- surface profile (form) of the cone (conicity);
- orientation and radial location of the cone (cone axis) relative to a datum;
- axial location of the cone relative to a datum;
- distance of an end face of the truncated cone relative to another face.

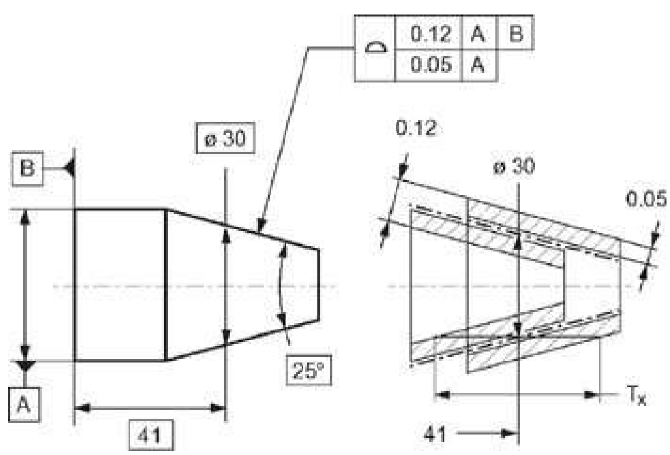
The first three characteristics (deviations) may be toleranced either together (in common) by the profile tolerance of the cone (t in Fig. 5.3), or separately by different tolerances (Figs 5.4 to 5.10).



$T_x = 0.02/\sin 12.5° = t/\sin \alpha/2$

$T_d/2 = 0.02/\cos 12.5° = t/\cos \alpha/2$

Figure 5.3: Cone tolerancing by: pro ile tolerance (0.02) of the cone theoretical exact cone angle (25) theoretical exact cone diameter (30) theoretical exact distance (41) of this cone diameter (30) from the datum B the axial location of the cone (30) is limited between T $_x$ / 2 from the distance (41) to the datum B the axis of the pro ile tolerance zone is coaxial to the datum A



$T_x = 0.12/\sin 12.5°$

Figure 5.4: Cone tolerancing by: pro ile tolerance (0.05) of the cone theoretical exact cone angle (25) theoretical exact cone diameter (30) theoretical exact distance (41) of this cone diameter (30) from the datum A the axial location of the actual cone diameter (30) is limited between T $_x$ / 2 from the distance (41) to the datum B the axis of the pro ile tolerance zone is coaxial to the datum A.

The cone angle may be specified as the theoretically exact angular dimension.